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Greater Noida, Uttar Pradesh

CivicSafe

Smart Civic Complaint & Public Safety Management System

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PRESENTATION ROADMAP

Agenda

1. Introduction & Problem Statement
2. Objectives & Scope
3. System Architecture & Modules
4. Database Design
5. Technology Stack
6. Methodology
7. Implementation & Screenshots
8. Results, Advantages & Future Scope

Introduction & Background

Civic issues such as uncollected garbage, plastic dumping, waste accumulation, water pollution, and public-safety hazards are a daily reality in most Indian towns and cities. Citizens have no simple way to report them — complaints go through phone calls, in-person visits, or scattered social media posts.

What CivicSafe does?

A full-stack mobile platform giving citizens a single app to photograph an issue, auto-tag its GPS location, classify it, and submit it directly to the responsible authority — while giving administrators a live map, a prioritised queue, and a full audit trail from first report to final resolution.

6
Incident Categories

<1 min
Guided Reporting Flow

98
Source Files

15,280+
Lines of Code

Problem Statement

1. Citizens do not know where or how to report a civic issue.
2. Manual complaint processes (calls, paper forms, visits) are slow and tedious.
3. Complaints lack precise location data, making field verification difficult.
4. Authorities cannot prioritise issues they cannot see on a map or dashboard.
5. There is little to no transparency for the citizen after a complaint is filed.

Objectives of the Project

1. Simple mobile flow to capture photo + GPS + timestamp, submit in under a minute.
2. Automatically route every submitted report to the correct government department.
3. Give administrators a live incident map, prioritised queue & analytics dashboard.
4. Maintain a full status-history timeline from Submitted to Resolved.
5. Support one-tap emergency dialling for time-critical safety situations.
6. Explore AI-assisted image classification to auto-suggest category & flag invalid photos.

Scope of the Project

Six Incident Categories (Citizen Reporting)

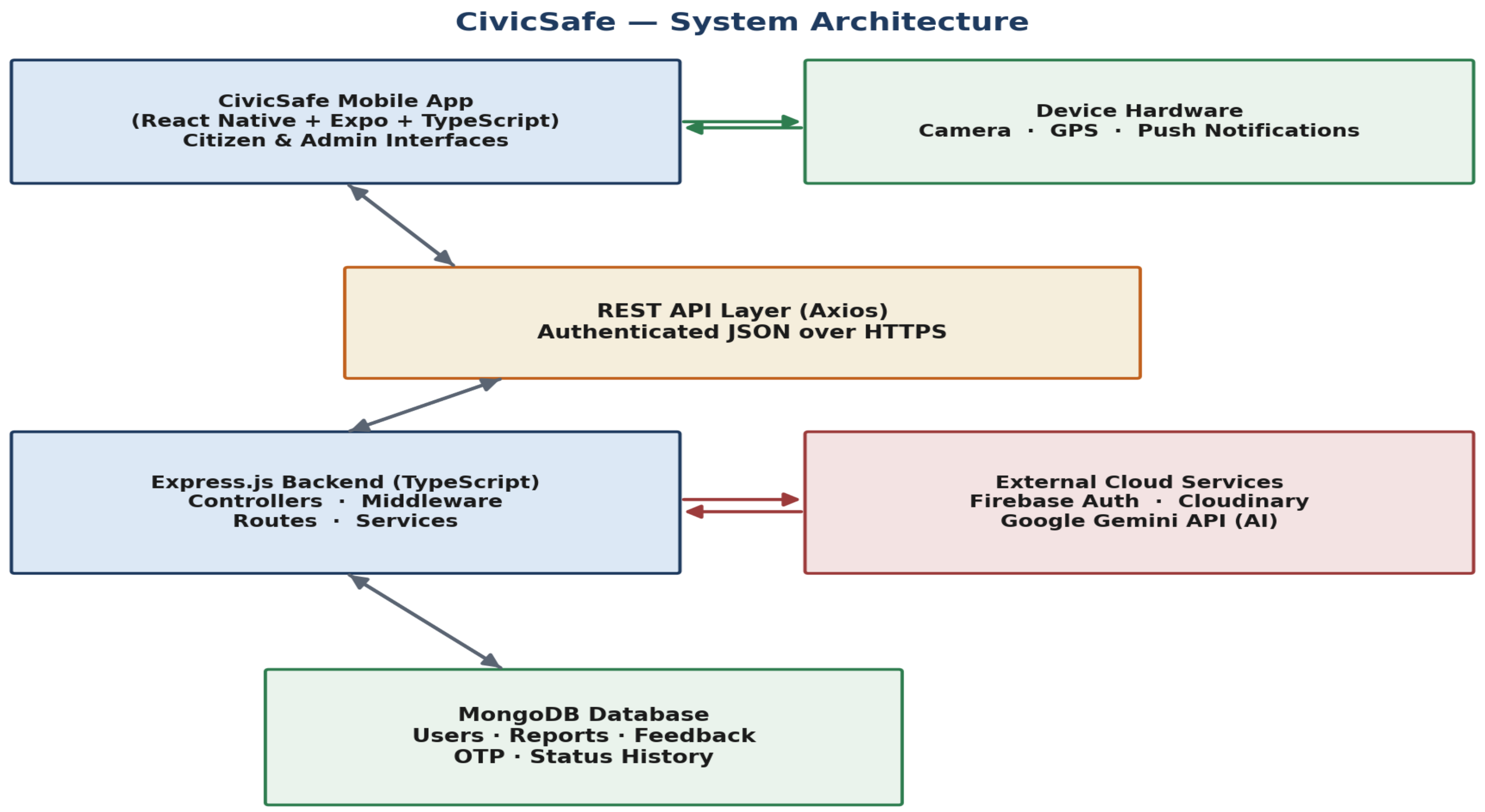
Garbage Dump	Plastic Pollution	Waste Accumulation
Water Pollution	Suspicious Object	Emergency Situation

Deployment Targets

Municipal Corporations	Smart-City Bodies	Pollution-Control Boards
Disaster-Management Cells	Police Departments	University Campuses

System Architecture

Three-tier architecture: React Native mobile client → Express.js REST API → MongoDB, supported by Firebase, Cloudinary & the Gem ini API.



Module Architecture — 8 Functional Modules

1. **Authentication & Onboarding**
   Role selection, registration, login, JWT session management

2. **Incident Reporting**
   Category, live camera capture, GPS/timestamp, description

3. **AI Detection & Validation**
   Gemini-powered category suggestion & image validation

4. **Report Tracking**
   Citizen view of live status across the report lifecycle

5. **Admin Dashboard & Analytics**
   Totals, pending/resolved counts, category trends

6. **Live Map**
   Interactive map of open incidents with directions

7. **Report Management & Resolution**
   Assign, update status, upload resolution proof

8. **Notification & Feedback**
   Push notifications + 1–5 rating and comment

Database Design — MongoDB Collections

Collection	Key Fields
User	name, phone, email, role, verificationStatus, department, fcmToken
Report	userId, category, description, imageURL, lat/long, timestamp, status, priority, assignedDepartment
StatusHistory	reportId, status, changedBy, remarks, changedAt
Feedback	reportId, userId, rating (1–5), comment
OTP	email, otp, expiresAt (TTL-indexed)

Technology Stack

Frontend / Mobile

React Native · Expo · TypeScript · React Navigation · Zustand · Axios

Backend

Node.js · Express.js · TypeScript · Helmet · CORS · Morgan

Database

MongoDB · Mongoose ODM

Authentication

JWT · Firebase Admin SDK · Role-based middleware

Media Storage

Cloudinary

AI Integration

Google Gemini API

Notifications

Push notification service (FCM token-based)

Email

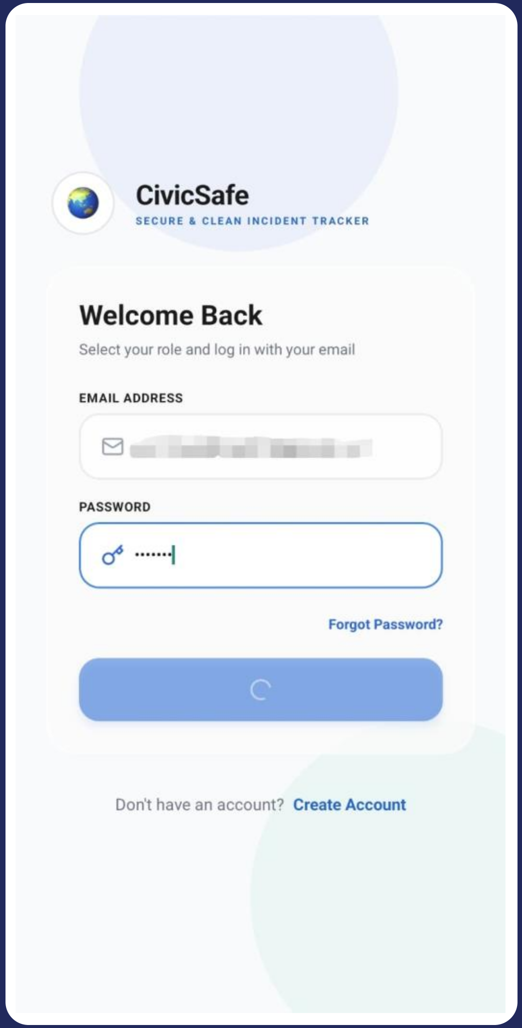
SMTP-based email service

Methodology

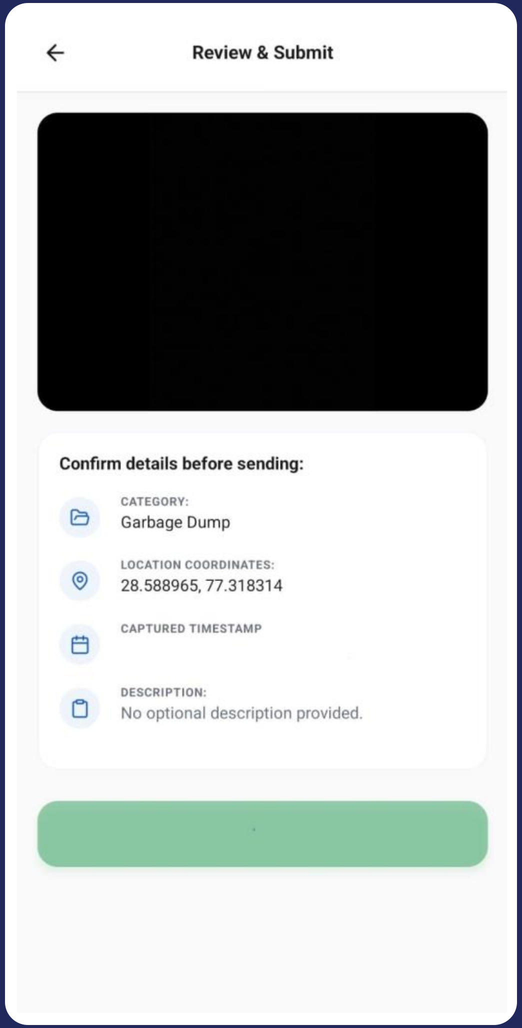
1. Requirement Gathering & Data Modelling
2. UI/UX Design
3. Backend Development
4. Frontend Development
5. Integration & Testing
6. Deployment & Demonstration

Thirteen guided backend steps collapse into six simple taps for the citizen — most reports are filed in under a minute.

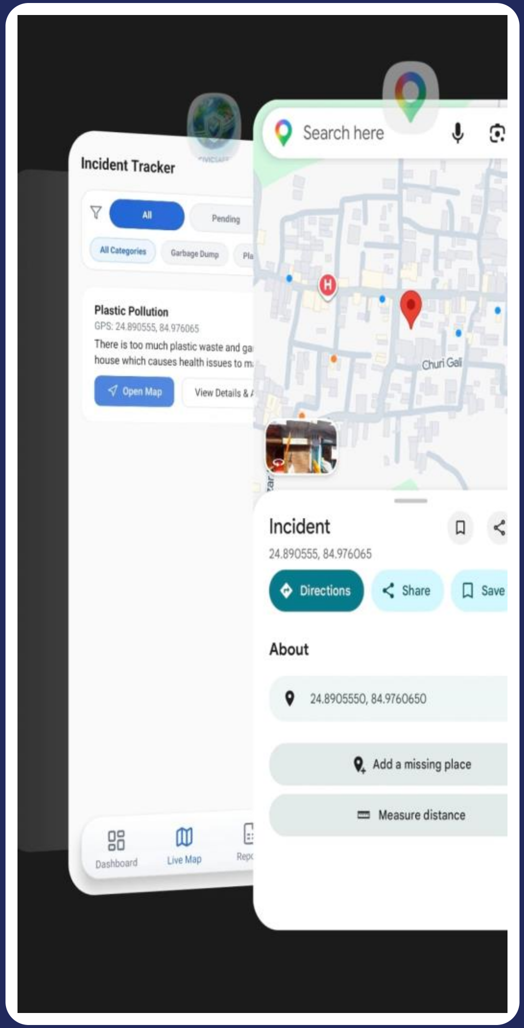
Application Walkthrough



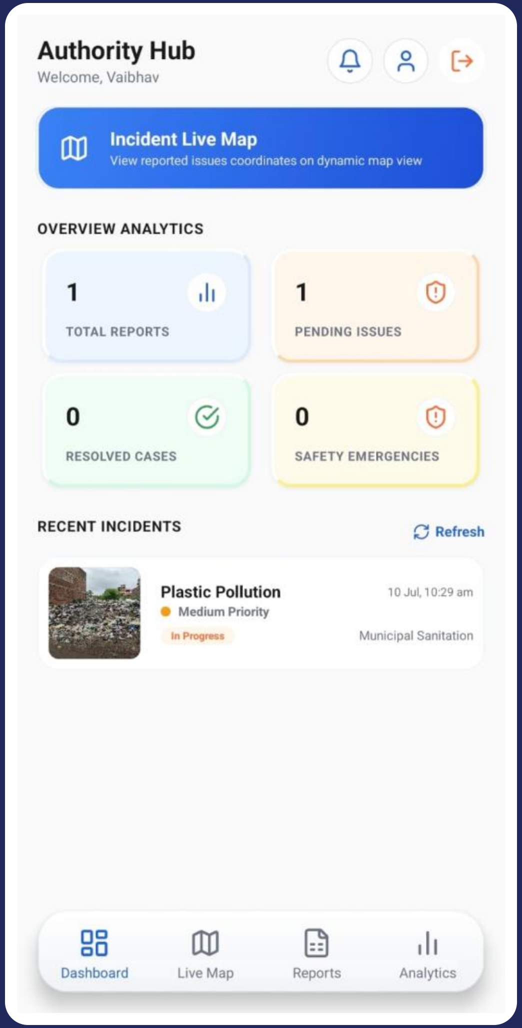
Login



Review & Submit



Live Map



Authority Hub

Results — Benefits to Citizens & Government

For Citizens

- Easy reporting, faster submission
- No paperwork or phone queues
- Transparent status tracking
- Confidence via photo + GPS evidence

For Government

- Centralised management of all incoming reports
- Real-time monitoring via the live map
- Data-driven prioritisation by category & volume
- Faster response and resolution times

Advantages & Applications

Advantages

- Photo- & GPS-verified reporting reduces false/vague complaints
- Guided six-tap flow keeps the citizen experience fast
- Automatic department routing removes manual triage
- Live map & analytics let officials prioritise with data
- Single-tap emergency dial for time-critical situations
- Full status-history timeline gives an auditable trail

Applications

- Municipal Corporations — sanitation & infrastructure complaints
- Smart Cities — integration into monitoring dashboards
- Pollution Control Boards — plastic & water pollution tracking
- Disaster Management — rapid citizen-sourced situational reports
- Police Departments — suspicious-object & safety intake
- University Campuses — scaled-down campus safety deployment

Limitations & Future Scope

Current Limitations

- AI layer runs as a client-triggered prototype, not a dedicated backend service
- Six fixed incident categories; taxonomy not yet user-extensible
- Routing rules are category-based, not yet configurable per city
- Load testing done at prototype/demo scale, not full production traffic

Future Scope

- Move AI inference to a dedicated backend model service
- Add multi-tenant support for multiple municipal bodies
- Expand language support beyond English/Hindi
- Add SLA-based escalation for stalled reports
- Integrate with existing municipal ERP/ticketing systems

From Prototype to Deployable Civic-Tech Platform

CivicSafe bridges the gap between citizens and government by turning GPS, camera, and role-based workflows into faster reporting, better monitoring, and real transparency for smart-city governance. It replaces an unstructured, low-visibility complaint process with a structured, geotagged, and auditable one — while keeping the citizen experience to a guided six-tap flow.

2

Coordinated Codebases

98

Source Files

15,280+

Lines of Documented Code

Thank You!